

1. Administrative & Study Identification

**Full Study Title:** A Phase 1, 2-Part, Open-label Study to Evaluate the Pharmacokinetics of Novel KarX (BMS-986519) and KarT (BMS-986520) Prototypes Versus the KarXT (BMS-986510) and KarX-EC (BMS-986519) Reference Following Single Doses, and to Explore the Effect of Food After Multiple Doses of Selected Prototypes in Healthy Adult Participants **Short Title/Acronym:** Novel KarX and KarT Prototypes PK Study **Protocol Number:** CN012-0043 (Other Identifier: U1111-1321-3511) **NCT Number:** NCT07063342 **Trial Status:** RECRUITING **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Xanomeline Enteric Capsule (KarX / BMS-986519), Trospium Chloride (KarT / BMS-986520), and Xanomeline/Trospium Chloride Capsule (KarXT / BMS-986510) **Phase of Development:** PHASE1 **Medical Monitor:** BMS Clinical Trials Contact Center

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate the pharmacokinetics (PK) of novel KarX and KarT drug prototypes versus the KarXT and KarX-EC reference following single doses, and to explore the effect of food on drug absorption after multiple doses of selected prototypes in healthy adult participants. **Secondary Objectives:** To evaluate the safety, tolerability, and optimal formulation viability of the novel formulations compared to the reference medicine.

**2.2 Background & Rationale** Xanomeline-trospium (KarXT) is a novel combination muscarinic agonist and antagonist utilized for treating psychiatric conditions such as schizophrenia. To optimize its delivery and patient experience, Bristol Myers Squibb is testing new formulations (KarX and KarT prototypes). This Phase 1 study is necessary to understand how the new prototypes' pharmacokinetic profiles compare against the reference formulations, and to observe if food intake significantly impacts the absorption of the medication over multiple doses.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Active Comparator (KarXT and KarX-EC reference formulations) **Blinding:** Open-label (None) **Randomization:** Randomized, Parallel Assignment

**3.2 Planned Enrollment & Duration Target** **\$N\$:** 48 healthy adult participants **Number of Sites:** Single-center / Multi-center clinical pharmacology units (e.g., Nottingham, UK) **Estimated Study Start:** 27-Jun-2025 **Estimated Primary Completion:** 15-Aug-2026

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**4. Subject Selection (Eligibility Criteria)**

**4.1 Inclusion Criteria** 1. Healthy volunteers between 18 to 55 years of age. 2. Body Mass Index (BMI) between 18.0 kg/m2 to 32.0 kg/m2, inclusive, at screening. 3. Willingness to comply with clinic confinement and protocol-defined procedures.

**4.2 Exclusion Criteria** 1. Presence of any unstable or significant medical, neurological, or psychiatric condition. 2. History of hypersensitivity to xanomeline, trospium chloride, or their excipients. 3. Other protocol-defined exclusion criteria limiting the safe participation in a Phase 1 PK trial.

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**5. Intervention & Treatment Plan**

**5.1 Investigational Regimen Dosage/Frequency:** Single oral doses of KarX, KarT, and KarX-EC prototypes vs reference formulations in Parts 1 and 2. An optional Part 3 includes multiple doses (twice a day for 21 days) of a selected prototype. **Route of Administration:** Oral (Capsules) **Duration of Treatment:** Up to 21 consecutive days for the multiple-dose cohort, requiring clinic confinement.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard Phase 1 allowed therapies (e.g., limited non-interfering medications approved by the investigator). **Prohibited:** Other investigational agents or medications known to interfere with the absorption or metabolism of xanomeline or trospium chloride.

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**6. Study Timeline & Visit Schedule**

|                         |               |                                                                                                        |
|-------------------------|---------------|--------------------------------------------------------------------------------------------------------|
| Visit ID                | Window (Days) | Key Activities/Procedures                                                                              |
| Screening               | Up to Day -1  | Informed Consent, Medical History, BMI verification, baseline safety labs                              |
| Intervention (Part 1/2) | Variable      | Single-dose administration, intensive PK blood sampling, safety monitoring                             |
| Intervention (Part 3)   | Days 1 to 21  | Multiple-dose administration (BID), food effect exploration (soft food/liquid vehicle for last 9 days) |
| End of Study            | Follow-up     | Final PK analyses, Vital Signs, Exit Safety Interview, Discharge                                       |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoints** 1. Maximum observed concentration (C<sub>max</sub>) 2. Time of maximum observed concentration (T<sub>max</sub>) 3. Area under the concentration-time curve from time zero to time of last quantifiable concentration (AUC(0-T))

**7.2 Secondary & Exploratory Endpoints** 1. Number of participants with Treatment Emergent Adverse Events (TEAEs) 2. Number of participants with Serious Adverse Events (SAEs) 3. Number of participants with Adverse Events of Special Interest (AESIs) Impact of soft food administration on the rate and extent of drug absorption.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous in-patient clinical monitoring, patient reports, vital signs, and ECGs. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal BMS clinical safety review team.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Pharmacokinetic (PK) Analysis Set (participants with evaluable PK data) and Safety Set (all participants who received at least one dose). **Sample Size Justification:** N=48 provides sufficient statistical power to observe differences in absorption rates and to map accurate PK curves across 3 experimental groups. **Handling of Missing Data:** Standard handling for PK modeling; missing data will generally not be imputed for primary PK outcomes.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Direct onsite monitoring at the clinical pharmacology unit to ensure strict adherence to dosing and PK sampling times. **Audit Trail:** eCRF entry tracking, data clarification forms, and laboratory sample custody logs.

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## 11. Study Identifiers & Governance

**Primary ID:** CN012-0043 **Registry Number:** NCT07063342 **Posting ID:** 810659662119946089 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate Novel KarX and KarT Prototypes Versus the KarXT and KarX-EC Reference Following

Single Doses, and to Explore the Effect of Food After Multiple Doses of Selected Prototypes in Healthy Adult Participants

**Official Regulatory Title:** A Phase 1, 2-Part, Open-label Study to Evaluate the Pharmacokinetics of Novel KarX (BMS-986519) and KarT (BMS-986520) Prototypes Versus the KarXT (BMS-986510) and KarX-EC (BMS-986519) Reference Following Single Doses, and to Explore the Effect of Food After Multiple Doses of Selected Prototypes in Healthy Adult Participants

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12. Clinical Framework (Adapted for PK Study)

**Primary Condition:** Healthy Volunteers **Diagnosis Threshold:** BMI 18.0 to 32.0 kg/m2, inclusive **Medical Methodology:** Pharmacokinetic evaluation and formulation testing **Optimization Target:** Drug absorption optimization and formulation food-effect evaluation

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13. Study Procedures & Methodology

**Management Pathway:** Clinical pharmacology unit standard operating procedures **Education Protocol (HCP):** Site initiation visits and Phase 1 trial protocol training **Education Protocol (Patient):** Informed consent review and clinic confinement briefing **Quality Audit Mechanism:** Independent sponsor auditing and PK sample tracking

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14. Observation & Follow-Up Schedule

**Total Duration:** Up to 13 weeks per participant **Onsite Visit Cadence:** Confinement to the clinical unit for up to 25 nights depending on cohort assignment **Remote Monitoring Frequency:** N/A (In-patient monitoring) **Checklist Review Interval:** Continuous clinical observation during confinement

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15. Sign-off & Approval

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|---------------------------------|--------|-----------|---------------|------|------|-------------|------|
| Role                            | Name   | Signature | Date          | :--- | :--- | :---        | :--- |
| <b>Principal Investigator</b>   | [Name] | _____     | [DD-MMM-YYYY] |      |      |             |      |
| <b>Clinical Operations Lead</b> | [Name] | _____     | [DD-MMM-YYYY] |      |      | <b>Lead</b> |      |
| <b>Statistician</b>             | [Name] | _____     | [DD-MMM-YYYY] |      |      |             |      |